Instructions:

1. Answer all questions on this paper. For multiple choice questions circle the letter of the **best** choice. For short answer questions, write your answer in the space provided.
2. No aids (such as calculators or language translators) are permitted.
3. Do NOT remove any pages from this paper.
4. You have 90 minutes to complete the exam. There is a total of 31 marks.

Part 1: Predict the output [10 x 1 marks]

In each row of the table below, mentally execute the code fragment on the left and enter the expected output in the box on the right. Each row is separate. Assume any necessary *import* statements have been done.

	Code Fragment	*Expected Output*
1.	print(2**3//2)	
2.	print(2.5 * 7.5 / 2.5)	
3.	print(20 % 7 * 7)	
4.	print(13 // 4 * 5. // 2)	
5.	numbers = [1,2,3,4,5,6,7,8] print(numbers[1:-1:3])	
6.	print(('a','b')+('b','a'))	
7.	print(list(range(8,0,-2)))	
8.	for ch in 'good': print(ch*2, end='')	
9.	a=1; b=2; c=3 print(a<b<c)	
10.	xx = [42, 'hi', 3.14] xx[0],xx[2] = xx[2],xx[0] print(xx)	

Use this area for scrap work.

Part 2: Circle the letter of the best answer. [5 x 1 mark]

11. Which of the following statements is **true**?

a) the same variable names may be used in different functions
b) a function must not call another function
c) a function must have at least one parameter
d) all function definitions must contain a return statement
e) each program must have a function named main

12. When calling the **cos** function from **math** which of the following statements is **true**?

a) The argument must be a variable.
b) The argument must be expressed in degrees.
c) The argument must be expressed in radians.
d) The argument cannot be an expression.
e) The argument must be an integer.

13. Which of the following functions does <u>not</u> require an import?

a) sqrt
b) exp
c) log
d) max
e) tan

11. Which of the following statements is **true**?

a) all elements in a list must be of the same type
b) the values of the elements in a list can be changed
c) it is not possible to take a slice of a slice of a list
d) an element of a list cannot be another list
e) every list has at least one element

12. Which of the following statements is **true**?

a) The arguments to the range function must be integers.
b) The sequence of numbers from the range function cannot be negative.
c) The sequence of numbers from the range function are always in ascending order.
d) The sequence of numbers from the range function always differ by 1.
e) The sequence of numbers from the range function always start at 0.

Use this area for scrap work.

The programming questions begin on page 4.

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Part 3: Programming [Total 16 marks]

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16. Complete the Python function that begins with the header shown below. This function computes an approximation to e^x using the equation given below (5 marks)

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

This function is given the values of *x* and *tolerance* as floats. Add terms to the summation as long as the absolute value of the current term is greater than the tolerance. Count the number of terms added to the sum. Return the sum of the series as the approximate value of e^x followed by the number of terms added to the sum.

Hint: Use floats to compute the approximate value of e^x.

```
def etox(x, tolerance):
```

17. Complete the Python function that begins with the header shown below. The function computes the X and Y coordinates of points along the trajectory of a projectile defined by the quadratic equation $y(x)=ax^2+bx+c$. The function is given a, b and c which are the coefficients in the quadratic equation, the horizontal distance (along the x-axis) the projectile travels and the number of intervals of equal width along the x-axis. Create a list of equally spaced X coordinates from x=0 to x=distance based on the number of intervals. Create a list of Y coordinates, one Y coordinate for each X coordinate, using the quadratic equation given above. Return the list of X coordinates followed by the list of Y coordinates. This function is used in question 18. (3 marks)

```
def computeTrajectory (a, b, c, distance, intervals):
```

18. Write a Python program that calls the functions *computeTrajectory* from question 17 to compute the trajectory of a projectile defined by the quadratic equation $y(x)=ax^2+bx+c$.
Do NOT rewrite *computeTrajectory* here! (8 marks)

- Input the integer number of intervals.
- Loop as long as the number of intervals is greater than zero.
- Input the launch angle of the projectile in degrees as a float.
- If the launch angle is less than or equal to zero display an error message that includes the launch angle as shown in the sample run of the program.
- Otherwise compute the values of a, b, c and distance using the following equations.

$$a = \frac{-G}{2v_0^2\cos(\theta)^2}\ ; b = \tan(\theta)\ ; c = y_0;\ distance = \frac{-b-\sqrt{b^2-4ac}}{2a}$$

where *G* is gravity at 9.81 m/s^2, *v0* is the initial velocity of 25.25 m/s, θ is the launch angle in degrees, y_0 is the initial height of 2.1 meters and distance is the horizontal distance the projectile travels in meters.

- Call compute Trajectory to get the list of X and Y coordinates.
- Display the heading shown in the sample run of the program.
- In a loop display the corresponding X and Y coordinates that define the points along the trajectory to 4 decimal places in 12 character positions, one pair of values per line of output.

Be sure to include any import statements required by the program.

A sample run of the program is shown below.

```
Enter the number of intervals (<= 0 to quit): 10

Enter the launch angle of the projectile (> 0): 0
The launch angle, 0, must be greater than zero!

Enter the number of intervals (<= 0 to quit): 10

Enter the launch angle of the projectile (> 0): 37.25
X coordinate Y coordinate
      0.0000       2.1000
      6.5277       6.5464
     13.0554       9.9580
     19.5831      12.3349
     26.1108      13.6771
     32.6385      13.9844
     39.1662      13.2571
     45.6939      11.4949
     52.2215       8.6980
     58.7492       4.8664
     65.2769       0.0000

Enter the number of intervals (<= 0 to quit): 0
```

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